

LAKAN SINGH DEGAVATH

Software & AI/ML Engineer | Deep Learning Inference | LLM Systems | Distributed Systems | Performance Optimization

United States | lakansingh53@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Software and AI/ML Engineer with 7 years of experience designing, developing, evaluating, and deploying production-grade machine learning and AI systems. Experienced with Python, NLP, LLMs, embeddings, transformers, PyTorch, TensorFlow, model inference, distributed systems, and scalable backend services. Built production AI applications and evaluation workflows with strong experience in model benchmarking, performance analysis, debugging, automated testing, monitoring, and reliability. Experienced in analyzing model behavior and evaluating ML and LLM approaches to build reliable, traceable, and production-ready AI solutions.

TECHNICAL SKILLS

Programming: C++, Python, C, Java, Scala, Go, SQL, JavaScript/TypeScript

Deep Learning & AI: PyTorch, JAX, TensorFlow, Keras, Hugging Face, Deep Learning Frameworks, Machine Learning, LLMs, Generative AI, NLP, scikit-learn, XGBoost

NLP & LLM Systems: Transformers, Embeddings, Semantic Search, RAG, LangChain, LangGraph, Hugging Face, Vector Databases, Prompt Engineering, Context Engineering

AI Evaluation, Safety & Reliability: LLM Evaluation, Model Evaluation, Evaluation Frameworks, Evaluation Pipelines, Model Benchmarking, Model Behavior Analysis, Failure-Mode Analysis, Edge-Case Testing, AI Safety, Trust & Safety, Compliance, Error Analysis, Model Monitoring, Automated Testing, Debugging, Production Reliability

LLM & Model Serving: LLM Inference, Model Serving, Inference Backends, vLLM, SGLang, Quantization, Batching, Low-Latency Inference, Performance Optimization

Inference & Systems: Deep Learning Inference, LLM Inference, Model Inference, Inference Backends, Model Serving, Low-Latency Systems, Distributed Systems, Concurrency, Asynchronous Programming, Performance Analysis, Performance Optimization, Fault Tolerance, High Availability.

Software Engineering: Python Development, REST APIs, FastAPI, Microservices, Software Design, System Design, Scalable Backend Services, Distributed Systems, Problem Solving, Debugging, Testing, Unit Testing, Integration Testing, Code Reviews, Git, Linux

AI Infrastructure: Docker, Kubernetes, MLflow, CI/CD, Experiment Tracking, Monitoring, Logging, Tracing, Observability

Distributed Computing: PySpark, Apache Spark, Spark Streaming, Kafka, Dask, Flink, Pandas, NumPy

Cloud: AWS, SageMaker, S3, EC2, Lambda

PROFESSIONAL EXPERIENCE

Naav Tech

AI/ML Engineer

United States · Feb 2026 - Present

- Evaluated and debugged production AI/LLM workflows through automated testing, edge-case testing, failure-mode analysis, performance analysis, monitoring, and systematic validation, identifying unexpected model behavior and reliability issues before production deployment.
- Architected and deployed production-grade Retrieval-Augmented Generation (RAG) applications using LangGraph, LangChain, OpenAI APIs, FastAPI, embeddings and vector databases, enabling low-latency enterprise knowledge retrieval.
- Designed, developed, and optimized low-latency model inference services and scalable inference backend components for real-time AI applications, applying distributed systems, concurrency, fault tolerance, debugging, and performance optimization techniques.
- Partnered with ML researchers, data scientists, and software engineers to productionize AI models through system design, code reviews, automated testing, debugging, performance analysis, and production optimization.
- Designed end-to-end AWS AI architectures using Amazon SageMaker, S3, Lambda, REST APIs, deployment, monitoring, and lifecycle management to support scalable production AI applications.
- Built reusable MLOps and LLMOps platforms using Docker, CI/CD, model versioning, monitoring, and experiment tracking to streamline AI development and deployment workflows.
- Optimized PySpark and streaming data pipelines supporting production machine learning workloads, improving scalability, reliability, and processing efficiency across distributed cloud environments.

Avila University

Graduate Research Assistant

United States · Apr 2024 - Dec 2025

- Built a reproducible Python NLP and machine learning pipeline for 205K+ product reviews, implementing data preprocessing, text normalization, feature engineering, model training, and evaluation for large-scale sentiment classification.

- Designed and executed model evaluation and benchmarking pipelines for Random Forest, Naive Bayes, and SVM using accuracy, precision, recall, and F1 score, establishing reproducible baselines for evaluating ML and LLM approaches across 205K+ product reviews.
- Evaluated GPT-4 through the OpenAI API for zero-shot and mixed-sentiment classification, analyzing model behavior, edge cases, prediction errors, and performance trade-offs against traditional machine learning baselines.
- Surveyed deep learning and NLP architectures including BERT, CNN/RNN/Bi-LSTM, Llama 2, Word2Vec, GloVe, FastText, and ELMo, and co-authored and presented the resulting research to faculty and peers.

HCLTech

Software Engineer

Hyderabad, India · Sep 2020 - Dec 2023

- Designed and built a production real-time machine learning inference platform processing 3M+ financial transactions, integrating model inference, distributed backend components, Kafka, and Spark Streaming into scalable production pipelines.
- Designed, analyzed, and optimized distributed streaming and model-inference pipelines, identifying and resolving performance bottlenecks to reduce near-real-time detection latency by 45%.
- Optimized machine learning models using ROC-AUC, Precision-Recall analysis, SHAP, and LIME, reducing false positives to fewer than 2 per 1,000 transactions while improving model explainability.
- Built reusable inference backend and distributed-processing components using PySpark and Dask, improving scalability, reliability, and operational efficiency at production scale.
- Conducted A/B testing comparing rule-based and machine learning-based fraud detection strategies, deploying the higher-performing solution and contributing to the prevention of over \$90K in annual fraud losses.
- Collaborated with data scientists, ML engineers, backend engineers, and product teams to productionize fraud detection solutions through system design, code reviews, automated testing, debugging, and production performance optimization.

Associate Software Engineering

Hyderabad, India · Sep 2019 - Aug 2020

- Developed and maintained C++, Java, and Scala components within a large enterprise codebase, applying object-oriented design, code reviews, automated testing, debugging, and production engineering practices to build reliable and maintainable system software.

PROJECTS

GPU-Accelerated LLM Inference & Performance Optimization

- Designed and implemented GPU-accelerated LLM inference pipelines using PyTorch, CUDA, TensorRT-LLM, and vLLM on NVIDIA GPUs, benchmarking latency, throughput, and GPU memory utilization across inference backends.
- Optimized TensorRT-LLM inference configurations through batching, quantization, and engine-build parameter tuning, analyzing latency, throughput, and GPU memory trade-offs for efficient LLM inference.
- Analyzed CUDA execution, GPU memory management, parallel computation, and kernel-level performance characteristics through close-to-metal performance analysis for hardware-aware deep learning inference optimization.
- Implemented GPU compiler optimization workflows using MLIR and LLVM, applying computational graph transformations and compiler passes to optimize execution paths for GPU-accelerated deep learning inference.

Enterprise Multi-Agent AI Platform

- Designed scalable LLM inference and retrieval services using Python, FastAPI, LangGraph, FAISS, and distributed backend components, supporting concurrent requests and low-latency enterprise knowledge retrieval.
- Built containerized model-serving APIs with Docker, monitoring, logging, automated testing, and CI/CD deployment on AWS, emphasizing reliability and production performance.
- Implemented modular retrieval and inference workflows supporting document ingestion, embeddings, context assembly, tool execution, routing, memory, and LLM response generation.

EDUCATION

Avila University

M.S., Computer Science – AI/ML Specialization

United States

- **GPA:** 3.91/4.0
- **Coursework:** Distributed Systems, Machine Learning, Artificial Intelligence, Cloud Computing, Data Structures & Algorithms, Computer Networks, Software Engineering, Big Data Analytics